MODULE 7

How Can You Be a Leak Detective and Water Saver?

WATER: NATURE'S PRECIOUS GIFT

TEACHING-LEARNING RESOURCE PACKAGE

Maynilad

Maynilad WATER ACADEMY
THE THOUGHT LEADER IN WATER AND SANITATION

UP NISMED 1964 - 2014

United Nations Educational, Scientific and Cultural Organization

Philippines National Commission for UNESCO

FPSMER Foundation for the Promotion of Science and Mathematics Education and Research, Inc.

SUGGESTED ENTRY POINT IN
THE K TO 12 CURRICULUM:

Grade 3 Mathematics (Measurement)

LEARNING COMPETENCIES:

- Converts time measure from days to weeks and months, and converts weeks, months and years to days

- Solves word problem involving time measure

NUMBER OF CLASS PERIODS: ❸

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Maynilad Water Academy
of Maynilad Water Services, Inc.

University of the Philippines
National Institute for Science and Mathematics
Education Development

Foundation for the Promotion of Science and Mathematics
Education and Research, Inc.

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PREFACE

Water is a precious gift from nature. The availability of water in sufficient quantity and acceptable quality to meet the needs of a population can spell the difference between a progressive and a deteriorating community. Water availability ensures the sustainability of natural environments. Water plays a crucial role in food and energy production, as well as in maintaining public health. It is considered central to social and economic development, as underscored in the United Nation's Sustainable Development Goals (SDGs). Access to water is considered a prerequisite for investment growth and poverty alleviation.

Recent weather events highlighted the impact of human activities on water sources. For the longest time, communities and industries have dismissed the importance of proper sewage disposal. Urban developers and construction companies have not carefully planned on water catchments. Inefficient agricultural activities have contributed to the deteriorating quality of water resources.

Every Filipino has the responsibility to protect and preserve our water resources for the future of our nation.

This teaching-learning resource package on water is a collaborative work by developers Maynilad Water Services, Inc. (Maynilad), the Foundation for the Promotion of Science and Mathematics Education and Research, Inc. (FPSMER), the University of the Philippines National Institute for Science and Mathematics Education for Development (UP NISMED), and the Philippines National Commission for UNESCO.

This package aims to equip educators with teaching processes and techniques that can help Filipinos understand the importance of water as key resource for attaining sustainable development, as aligned with Education for Sustainable Development or ESD objectives. The listing of titles suggests the sequence of use from the simplest to the most complex topics about water. The modules are intended to supplement the Philippines' K to 12 curriculum, with a K to 12 entry point provided for each title. Aside from schools, the package may also be used by government and non-government agencies, local authorities and volunteer groups for community education efforts and to contribute to local and national scientific data bases, as appropriate.

This package is a work-in-progress. Comments and suggestions for improvement are welcome. The use of the materials is highly encouraged to support the developers' shared mission to educate the public about water and environmental sustainability.

Content of the modules may be used and/or adapted to other platforms, partially or in full, with proper acknowledgment of the developers and the package as source material. Adoption of the modules entails no cost, but permission should be secured from:

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ABOUT MAYNILAD

Maynilad Water Services, Inc. (Maynilad) is the water and wastewater services provider for the 17 cities and municipalities that comprise the West Zone of the Metropolitan Manila area. It is an agent and contractor of the Metropolitan Waterworks and Sewerage System (MWSS).

In 1997, Maynilad was granted a 25-year exclusive concession by the Philippine Government to operate, maintain and invest in the water and sewerage systems in the cities of Manila (all but portions of San Andres and Sta. Ana), Quezon City (west of San Juan River, West Avenue, EDSA, Congressional, Mindanao Avenue, the northern part starting from the Districts of Holy Spirit and Batasan Hills), Makati (west of South Super Highway), Caloocan, Pasay, Parañaque, Las Pinas, Muntinlupa, Valenzuela, Navotas and Malabon, all in Metro Manila; the cities of Cavite, Bacoor and Imus, and the towns of Kawit, Noveleta and Rosario, all in the Province of Cavite. The concession term was extended by 15 years in 2010, after Maynilad was required by the Philippine Government to increase and accelerate its wastewater investments.

Module 7

How Can You Be a Leak Detective and Water Saver?

CONTENTS

Module 7

How Can You Be a Leak Detective and Water Saver?

A. OBJECTIVES

- Detect leaks at home
- Identify different ways of saving water

B. LEARNING ACTIVITIES

Activity 1: How do I know that there are leaks at home?

What you need:

- Food color or dye powder
- Clock or watch
- Toilet
- Bottle for collecting water

What to do:

1. Turn off all faucets in the house. Check the water meter. If the dial of the water meter continues to move, then there is a leak in the house. Proceed to Step 2. If the dial is not moving, then there is no leak.
2. Check for leaks in the toilet.
    - Close all faucets, stop using a hose and/or appliances (e.g., washing machine) that use water.
    - Go to the toilet. Ask an adult to help remove the lid off the toilet tank to prevent breaking it. (Note: The lid is heavy and hard to move).
    - Add a few drops of food color or a pinch of dye into the tank. DO NOT flush the toilet.
    - Wait for 15 minutes. Check if color appears in the toilet bowl without flushing the toilet. If it does, the toilet tank has a leak.
    - Flush the toilet immediately after the activity to avoid stain ing inside the tank.

Q1. A leaking toilet can waste about 750 liters of water every day. How many liters will a leaking toilet waste in a week? Show how you got the answer.
Q2. Some soft drink bottles marked Litro have a 'one liter' capacity. How many Litro bottles of water are wasted in one day if the toilet is leaking? In a week? Show your work.

2. Measure the volume of water wasted from a leaking faucet. (Note: If there is no leaking faucet, simulate one by slightly opening the faucet until water drips from it.)

- Get an empty 1-L soft drink bottle. Volume of the bottle: ____________
- Collect water from a leaking faucet using the bottle. Number of bottles of water collected for 30 minutes: ______

Q3. What is the capacity of the bottle that was used for col lecting water from the leaking faucet?
Q4. How many bottles of water are collected from the leaking faucet in 30 min? In one hour? In one day?
Q5. If the faucet is not repaired and it continues to leak for one day, how many liters of water will be wasted?
How many soft drink bottles of water are wasted in one day if the toilet is leaking? In a week? Show your work.

2. Measure the volume of water wasted from a leaking faucet.

- Get an empty bottle. Note the capacity or volume of the bottle in liters. Volume of the bottle: ____________
- Collect water from a leaking faucet using the bottle. Number of bottles of water collected for 30 minutes: ______

Q3. What is the capacity of the bottle for collecting water from the leaking faucet?

Q4. How many bottles of water were collected from the leaking faucet in 30 min? one hour? one day?

Q5. If the faucet is not repaired and it continues to leak for one day, how many liters of water will be wasted due to the leaking faucet?

Activity 2: What are some ways to save water?

What you need:

- Checklist of practices

What to do:

1. Read the list about practices in using water in and out of the home. Check the proper column if you agree or disagree that each practice is a way of saving water.

Checklist of Practices

Checklist of Practices

No.	**Do we save water if we do the following?**	Agree	Disgree
1.	Monitor your water bill if water use is unusually high.		
2.	Water the lawn/garden in the early morning or evening.		
3.	Wash fruits and vegetables in running water.		
4.	Use a tabo (dipper), pail and broom to clean the driveway and sidewalk.		
5.	Designate one drinking glass for each family member each day to cut down on the number of glasses to wash.		
6.	Turn off faucets tightly after each use.		
7.	Keep water running while brushing the teeth.		
8.	If ice cubes are accidentally dropped when filling a glass with ice, do not throw the ice in the sink. Place them in a pot with a plant.		
9.	When washing your hands, close the faucet while applying soap to your hands.		
10.	If your toilet does not have a half-flush option, place a one-liter plastic bottle with water in the toilet tank.		
11.	Check toilet tanks regularly for leaks.		
12.	Use rinse-water after laundry to water the plants.		
13.	When washing dishes, rinse them in running water.		
14.	Wipe oily and dirty plates with cloth or tissue before washing them.		
15.	Wash cars in the lawn using mild detergent and a pail of water instead of a hose.		

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16.	Remove weeds in the garden; they compete with water intended for plants.		
17.	Leave the reading of the water meter to the water provider personnel.		

Q1. How many items did you check under column Agree? Under column Disagree?

Q2. In a group of five, discuss the result of the survey. Which item did all your group members have the same response? Which items were not agreed on by most students? Discuss the items not agreed on by all members of the group.

Q3. There are 4 items in the list that do not help save water. These items are 3, 7, 13, and 17. Choose one of the items nd discuss why this practice does not help in saving water.

Assignment:

Ask the pupils to plan how to save water in their homes.

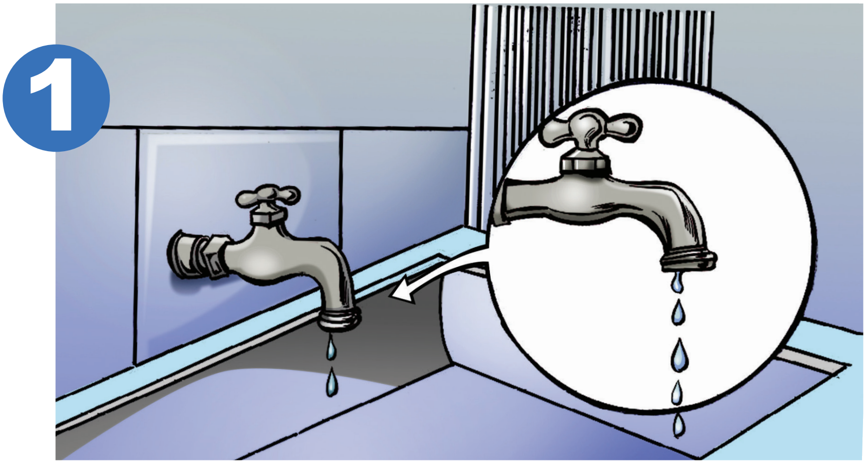
Activity 3: Who repairs a leaking faucet?

What you need:

•Pictures of activities

What to do:

1.Study the pictures. The pictures tell a story.
2.Tell what is happening in each picture.



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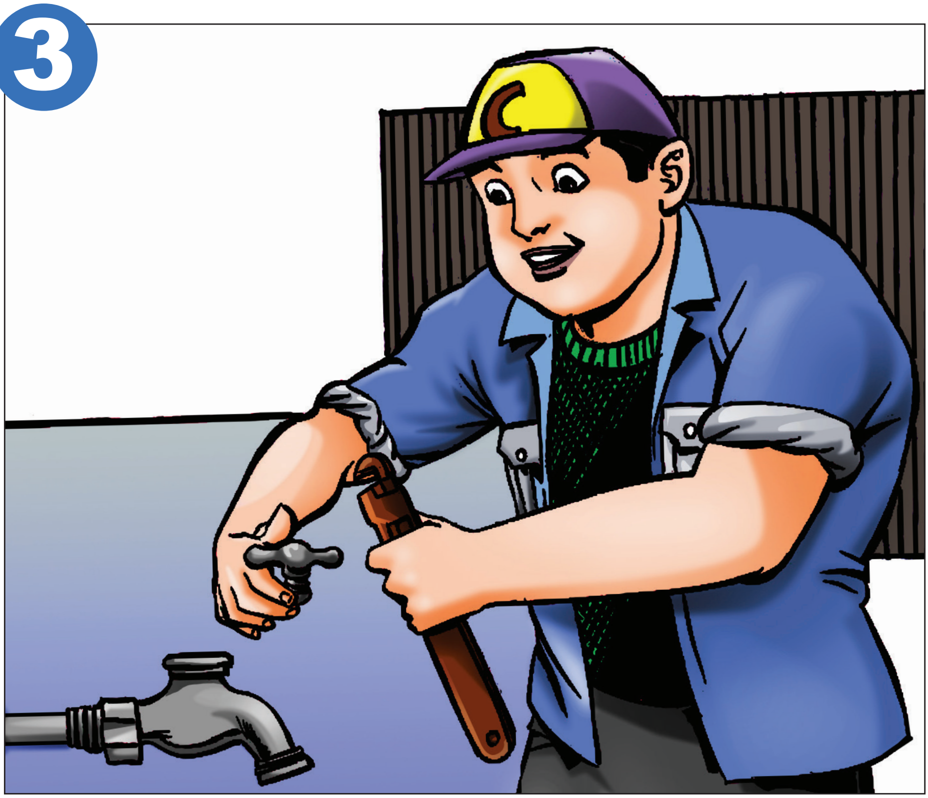

Q5. What is the job of the person referred to in the picture?
Clue: 7-letter word
Q6. Do you know someone in your community who doesthis work? Name him or her.
Q7.Why is that person important in the community?

C. TEACHER'S NOTES

C.1 How to use the activities

Activity 1 teaches children how to check for water leaks. It also helps them to know how much water is wasted due to a leaing toilet/faucet.

Note: ***Set an unopened bottle of 1-L soft drink. Mark the level of the liquid with a marking pen. The mark shows 1-L of liquid. Use the marked bottle as basis for marking an opened bottle to show a volume of 1-L.***

Activity 2 enables children (and their parents) to look into their prac-tices to save water.

Activity 3 introduces a plumber and the services that he/she offers to the community.

C.2 Answers to questions

Activity 1. How do you know that there are leaks at home?

Q1. About 750-L of water is wasted when the toilet is leaking.
To compute for water wasted in a week:

750-L/day x 7 days/week = 5,250-L per week

Q2. One soft drink bottle has a capacity of 1-L. A leaking faucet wastes 750-L = 750 bottles.
In a week,
750 bottles/day x 7 days/week = 5,250 bottles per week

Q3. 1-L
Q4 and Q5. Answers vary according to the degree of leakage.

Activity 2. What are the ways to save water?

Sample data:
•Volume of bottle collector (a)
•Number of bottles of wastewater collected in 30 min (b)
•Number of bottles of water collected (a x b) x 2
•Amount of water in one day if faucet is not repaired:
(a x b) x 2 x 24 hours

Activity 3. Who repairs a leaking faucet?

Q1. Picture 1: leaking faucet
Q2. Picture 2: man using tools to repair the faucet
Q3. Picture 3: man putting Teflon tape
Q4. Picture 4: man putting back the faucet; faucet no longer dripping
Q5. A plumber
Q6. Answers will vary.
Q7. Not many people know how to fix a leaking faucet. A professional plumber helps save water if there are leaking faucets or pipes.

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LIST OF TITLES IN THE PACKAGE